

Stage 19: Vendor Settlements, Payouts, Ledger & Financial Reconciliation

1. Architectural Overview

The **Vendor Settlements, Payouts, Ledger & Financial Reconciliation Engine** provides automated escrow settlement, statutory tax withholding (Section 194-O TDS & Section 52 GST TCS), commission invoicing, payout withdrawal pipelines, and an immutable double-entry wallet transaction ledger for the Alight International Multi-Vendor Marketplace.



2. Core Entities & Database Structure

(V45__vendor_settlements_payouts_and_ledger_enhancements.sql)

2.1 Vendor Wallets (vendor_wallets)

- pending_balance : Funds held in escrow while orders are being prepared, shipped, and delivered.
- available_balance : Settled funds eligible for payout withdrawal.
- total_earnings & total_withdrawn : Cumulative lifecycle accounting totals.
- total_commission_paid & total_tcs_paid : Deductions tracked for GST and IT filing.

- `bank_account_number` , `bank_ifsc_code` , `bank_name` : Verified payout disbursement details.

2.2 Wallet Transaction Ledger (`wallet_transactions`)

- `transaction_type` : `ESCROW_HOLD` , `ESCROW_RELEASE` , `COMMISSION_DEDUCTION` , `TCS_DEDUCTION` , `REFUND_REVERSAL` , `PAYOUT_DEBIT` , `ADJUSTMENT` .
- `amount` & `balance_type` : `PENDING` vs `AVAILABLE` .
- `balance_after` : Snapshot of wallet balance post-transaction for financial audit integrity.
- `reference_id` : Sub-order number or payout reference.

2.3 Vendor Payouts (`vendor_payouts`)

- `payout_reference` : Reference code (e.g., `PAY-2026-XXXXXXX`).
- `amount` & `status` : `PENDING` , `PROCESSING` , `PAID` , `REJECTED` , `CANCELLED` .
- `utr_number` : Bank Unique Transaction Reference recorded upon disbursement.
- `rejection_reason` & `admin_notes` : Complete audit trail for compliance.

2.4 Tax Compliance & Commission Invoices (`marketplace_commission_invoices` & `tax_compliance_ledgers`)

- Monthly GST commission bills with CGST/SGST/IGST breakdown and SAC code 998311.
- Quarterly GSTR-8 return summaries for marketplace operators.

3. Financial Settlement Mechanics

When a sub-order is marked `DELIVERED` :

1. **Gross Escrow Cleared:** Full gross amount is debited from `pending_balance` .

2. **Platform Commission Deduction:**

$$Commission = Subtotal \times \frac{Vendor\ Commission\ Rate}{100}$$

3. **Statutory 1% GST TCS Deduction:**

$$TCS = Subtotal \times 0.01$$

4. **Net Vendor Credit:**

$$Net\ Credit = Gross\ Amount - Commission - TCS$$

5. **Ledger Entries:**

- `ESCROW_RELEASE` (Credit to Available)
- `COMMISSION_DEDUCTION` (Commission Debit)
- `TCS_DEDUCTION` (TCS Debit)

4. Verification & Integration Suite

Verified via `SettlementAndPayoutsIntegrationTest.java` :

- ✓ **Escrow Hold & Release Flow** : Gross escrow holding upon order creation, statutory 1% TCS and 10% commission deductions upon delivery, net available balance credit, and 3 corresponding ledger entries.

- ✓ **Refund Reversals:** Clean escrow deduction upon order cancellation.
- ✓ **Bank Details Configuration:** Bank account, IFSC, holder name, and branch configuration.
- ✓ **Payout Request & Admin Approval:** Balance debit, UTR recording, and **totalWithdrawn** progression.
- ✓ **Payout Rejection & Balance Restoration:** Available balance restoration with **ADJUSTMENT** ledger entry.
- ✓ **Payout Validation Rules:** Guarding against insufficient funds or unconfigured bank accounts.
- ✓ **Admin Financial Metrics** : Total escrow, available balance, commissions collected, TCS collected, and disbursed totals.
- ✓ **Tax Compliance & Invoicing** : Monthly commission invoice generation and GSTR-8 quarterly summary generation.